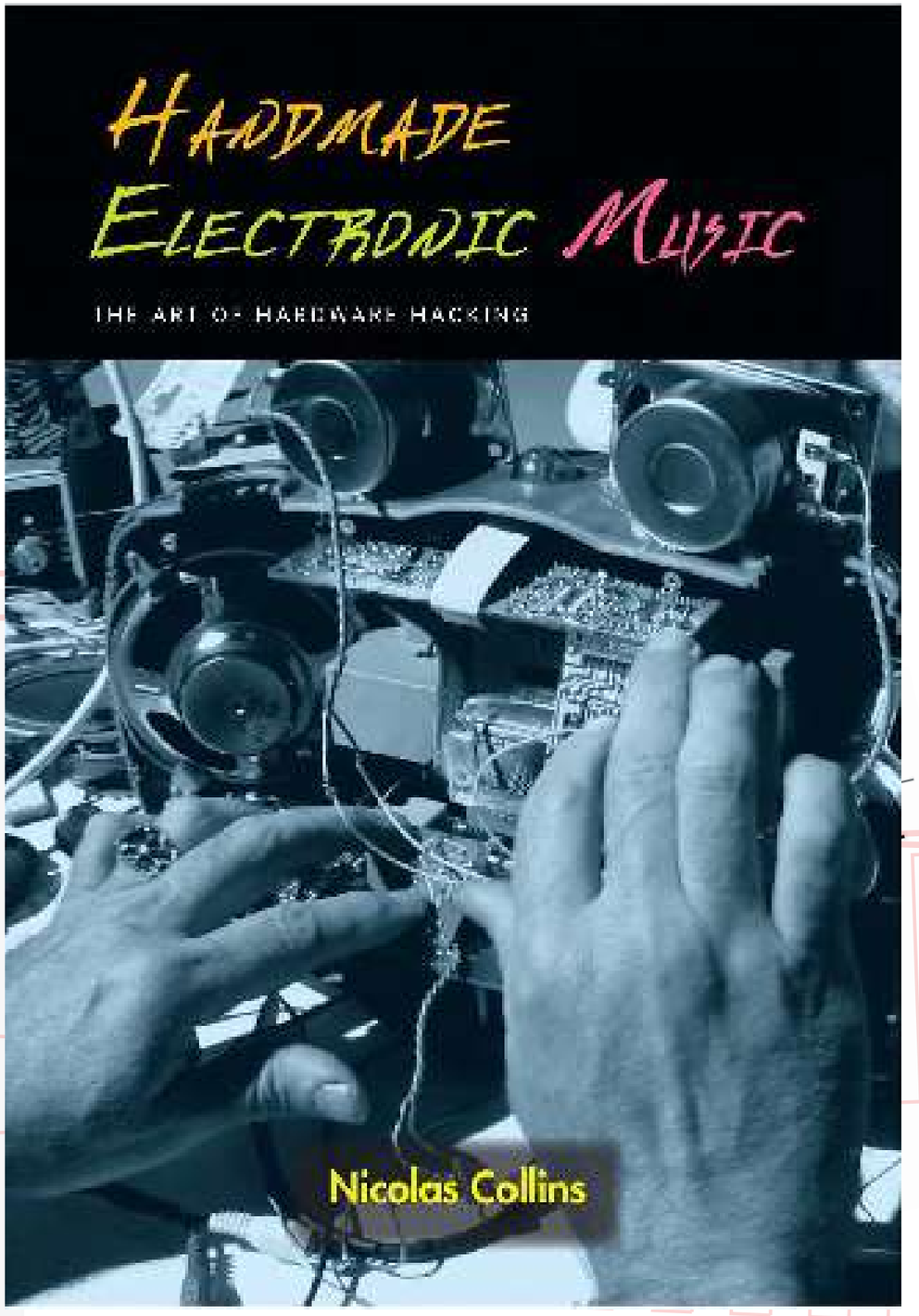
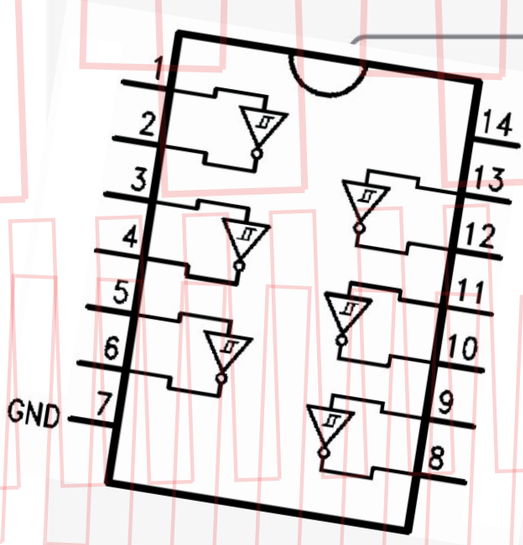
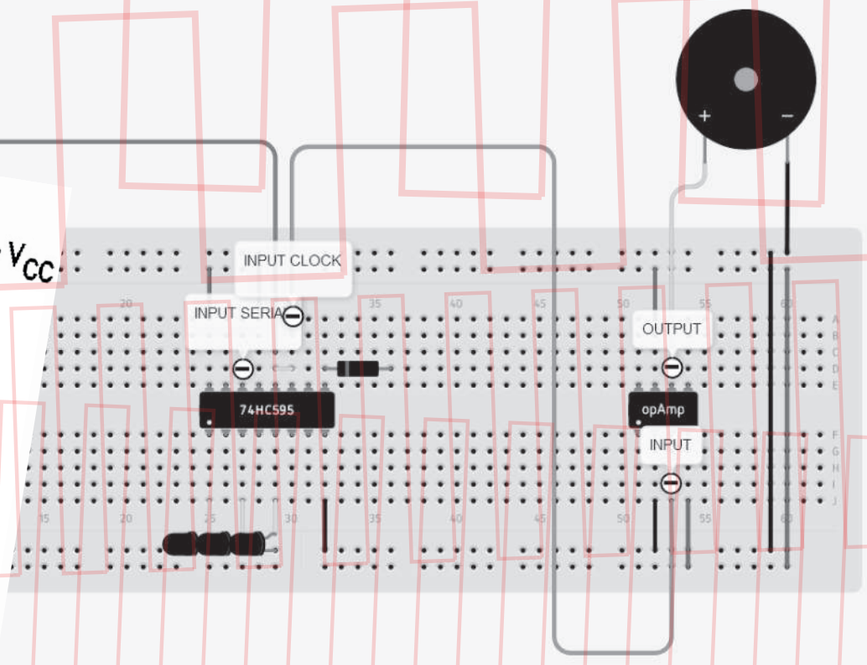
LOGIC NOISE WORKSHOP !

WE ARE GOING TO MAKE
OUR VERY OWN
ELECTRONIC MUSIC FROM
SIMPLE LOGIC GATES !



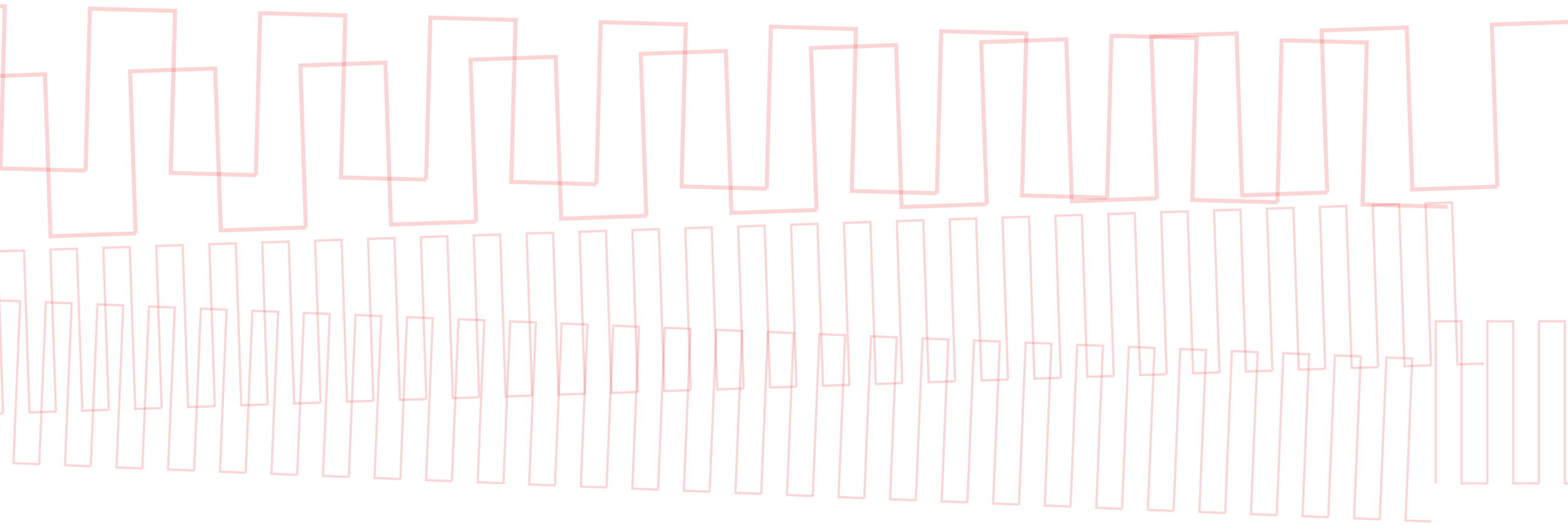




LOGIC NOISE WORKSHOP !

Learn to :

use an oscilloscope
prototype on a breadboard
wire up 74* logic chips
amplify, mix and modulate signals



LOGIC NOISE WORKSHOP !

Groups of 2

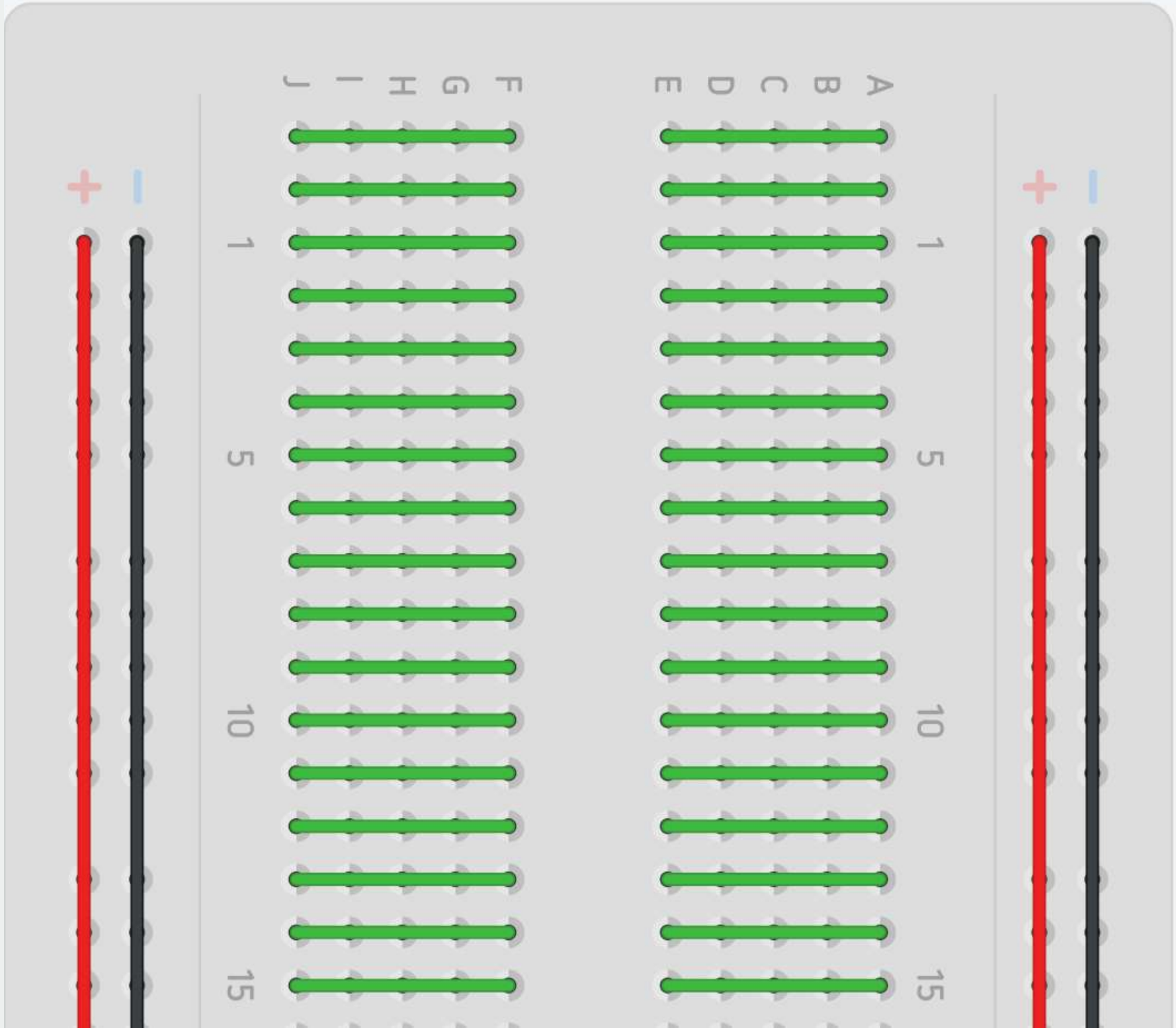
1x breadboard + wires
1x 5V power supply
1x 74HC14 hex inverter chip
1x CD4040 binary counter
1x LM386 audio amp chip
1x red LED
1x green LED
1x potentiometer
4x capacitor (0.01uF-10uF)
4x resistors (330Ω - 10KΩ)

breadboard connections

the convention is to use RED for + and BLACK or BLUE for -

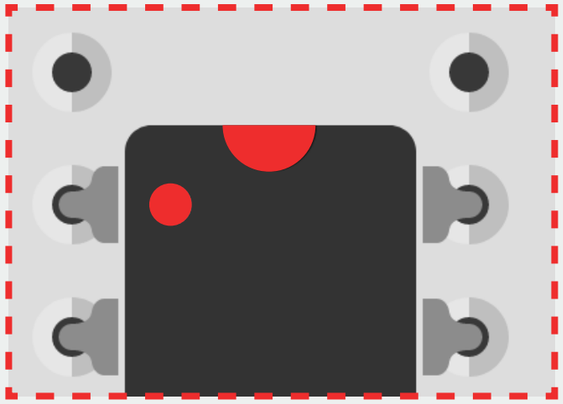
*note that the columns on the left and right are not connected together !

*note also that the top and bottom halves of your board may not be connected !

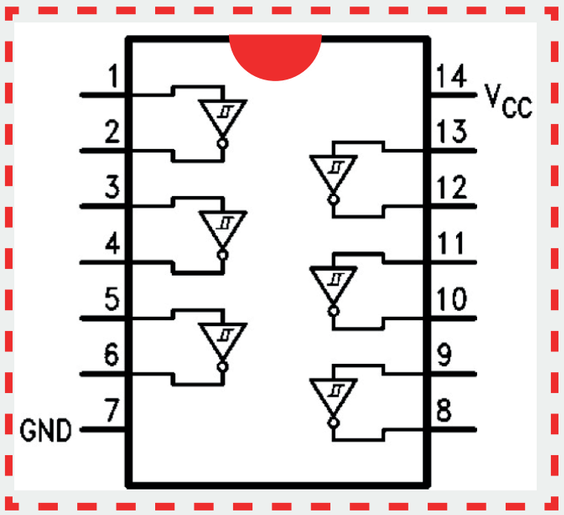


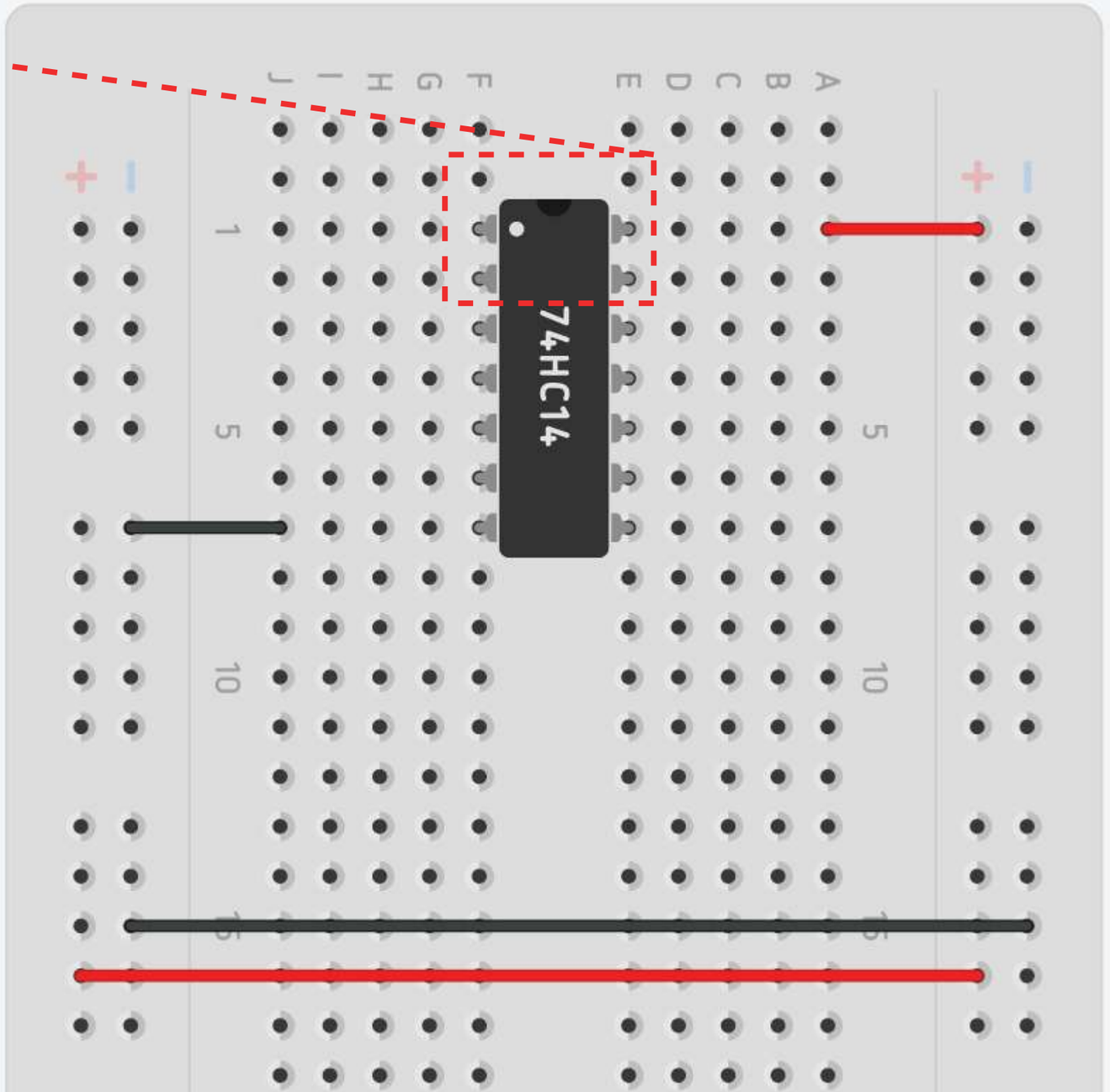
step 1

74HC14 has six inverters



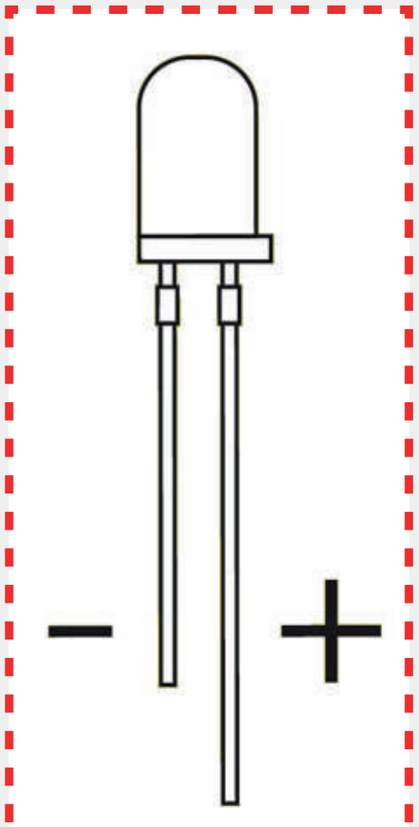
*notch should face north!



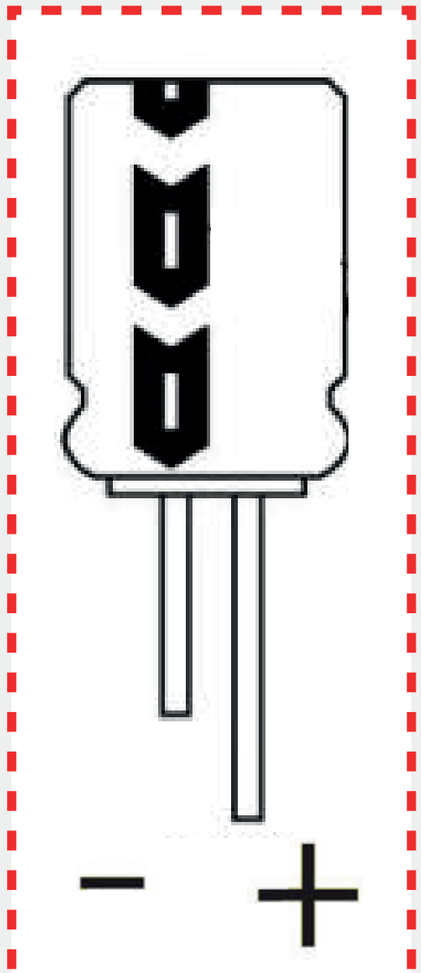


step 2

capacitors store tiny amounts of electricity while LEDs convert it to light !

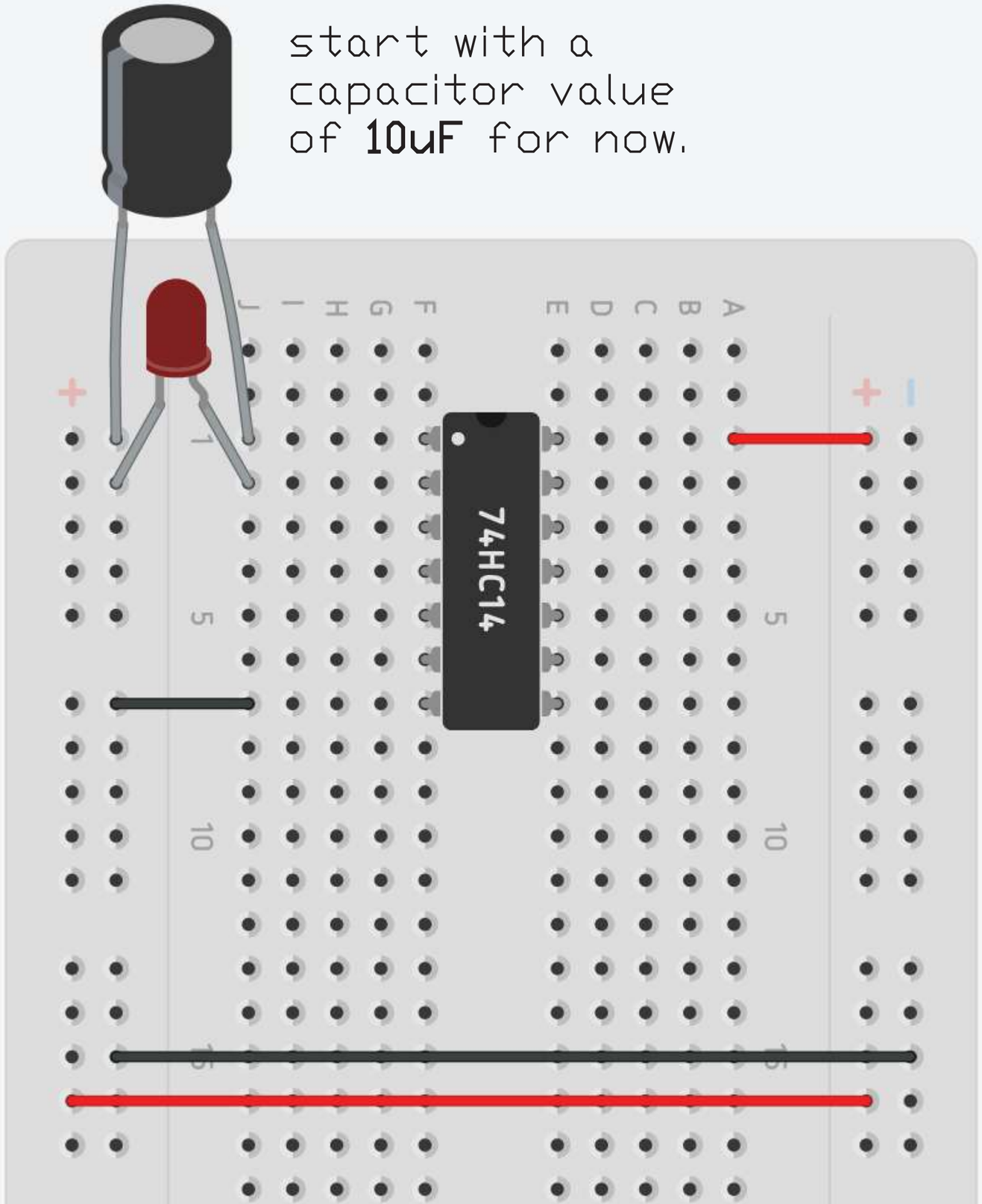


*LED polarity

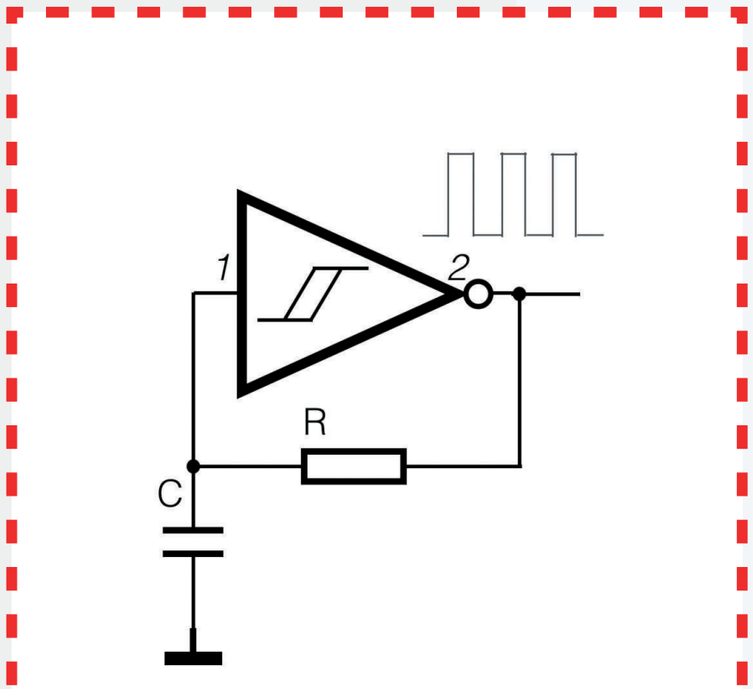


*capacitor polarity

start with a capacitor value of **10uF** for now.

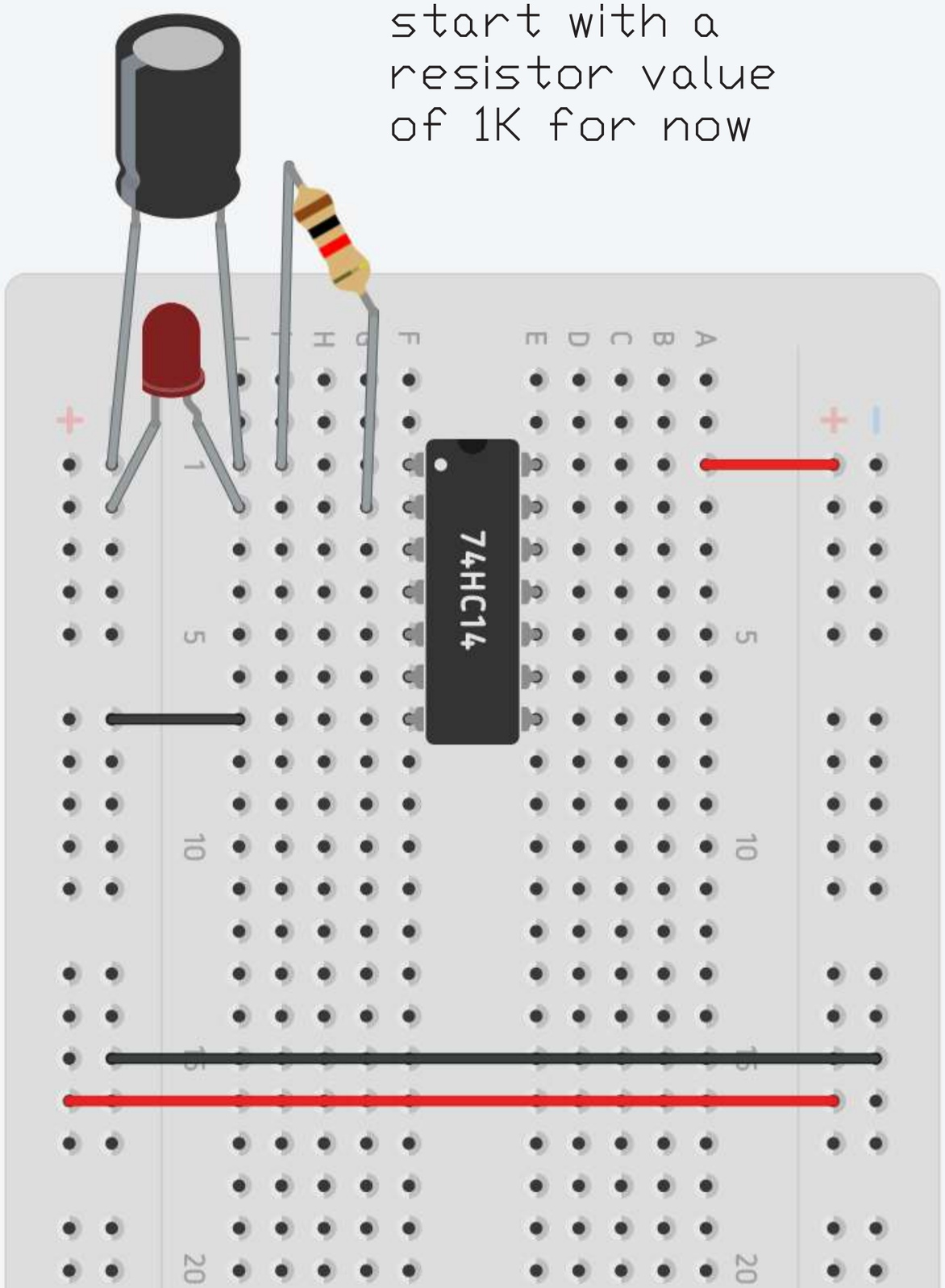


step 3



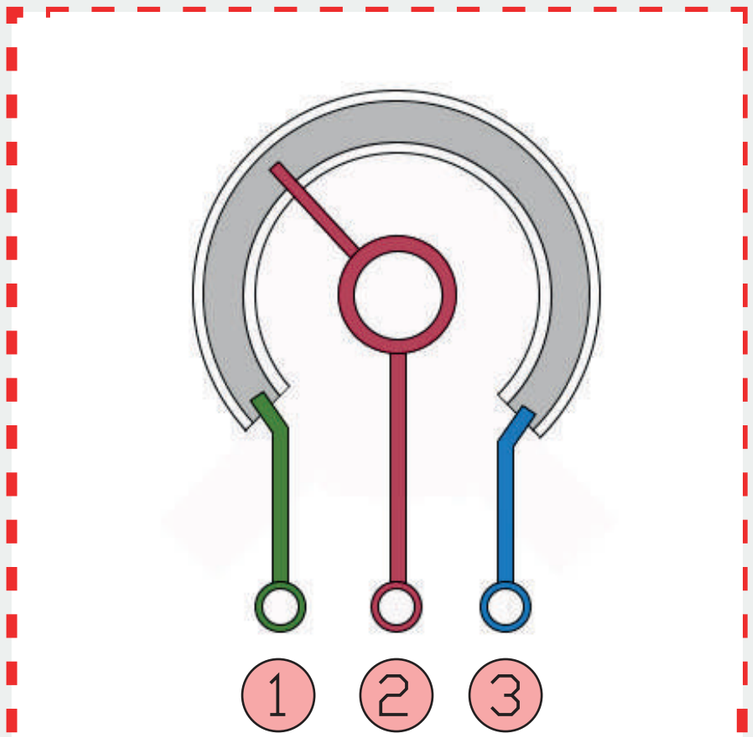
this is the
equivalent circuit

start with a
resistor value
of 1K for now



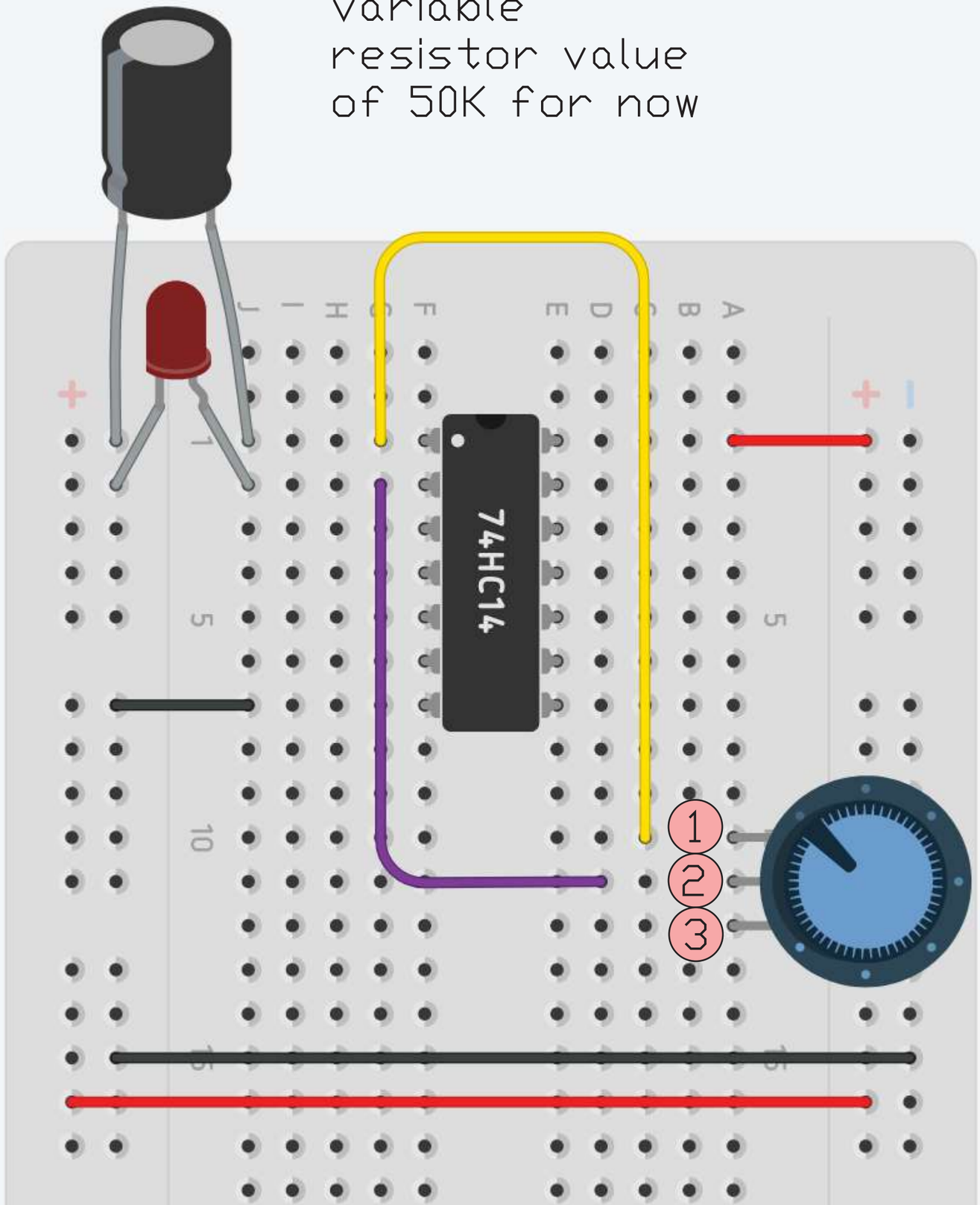
step 6

A potentiometer is a variable resistor



*potentiometer polarity

start with a variable resistor value of 50K for now

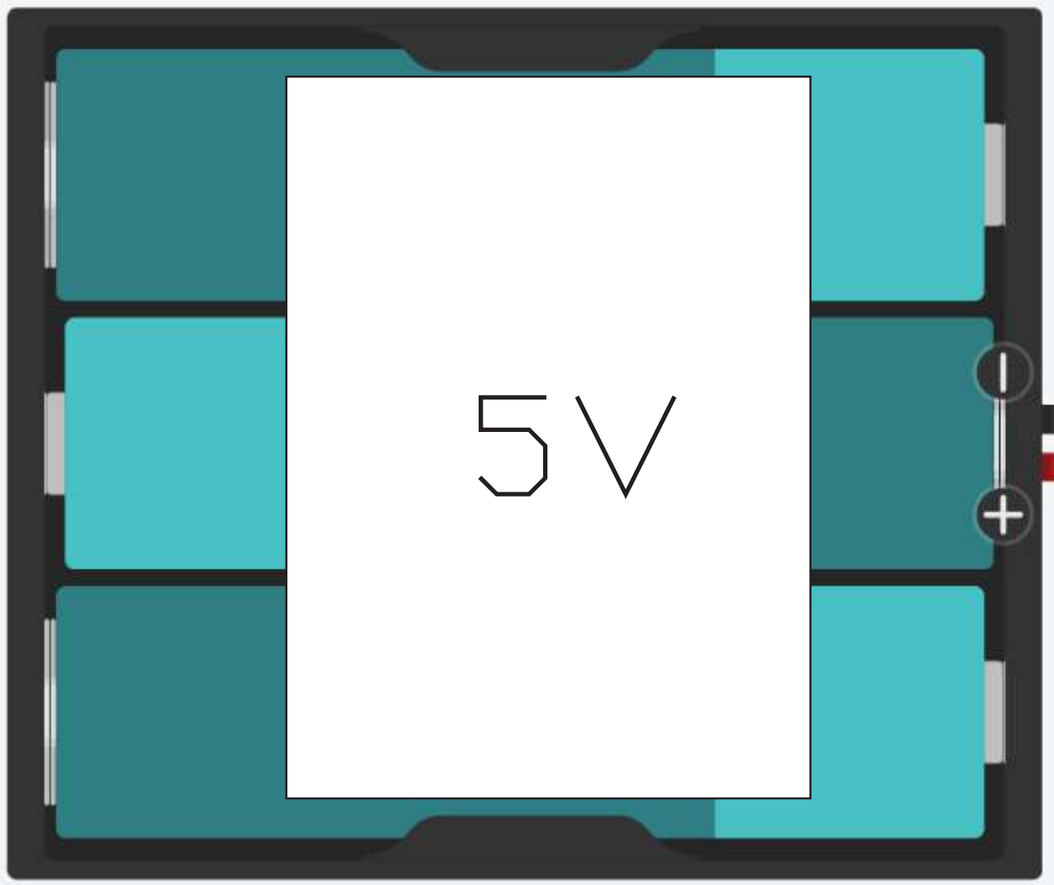


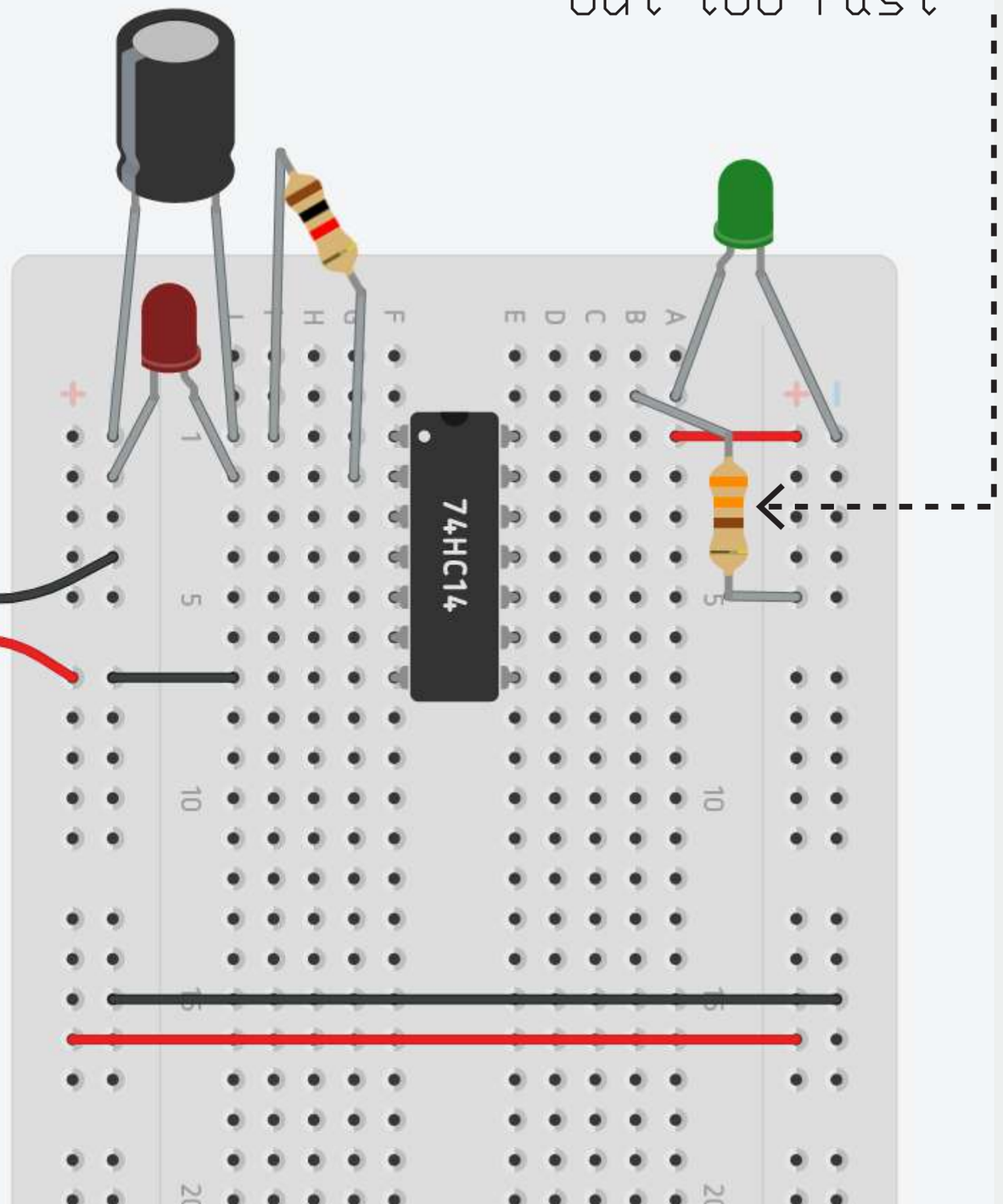
step 4

this 330 ohm resistor protects the LED for burning out too fast

*make sure to avoid a **short circuit** by connecting + to - !

*make sure to avoid a **reverse polarity** by connecting + to - !





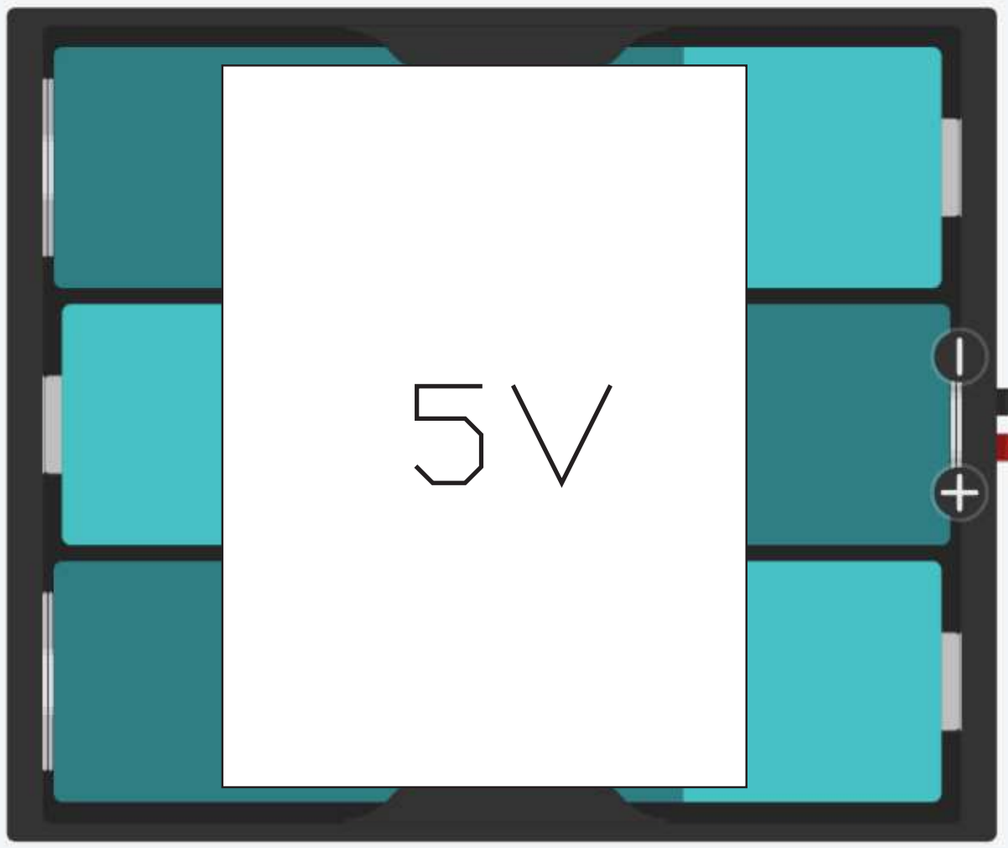
step 5

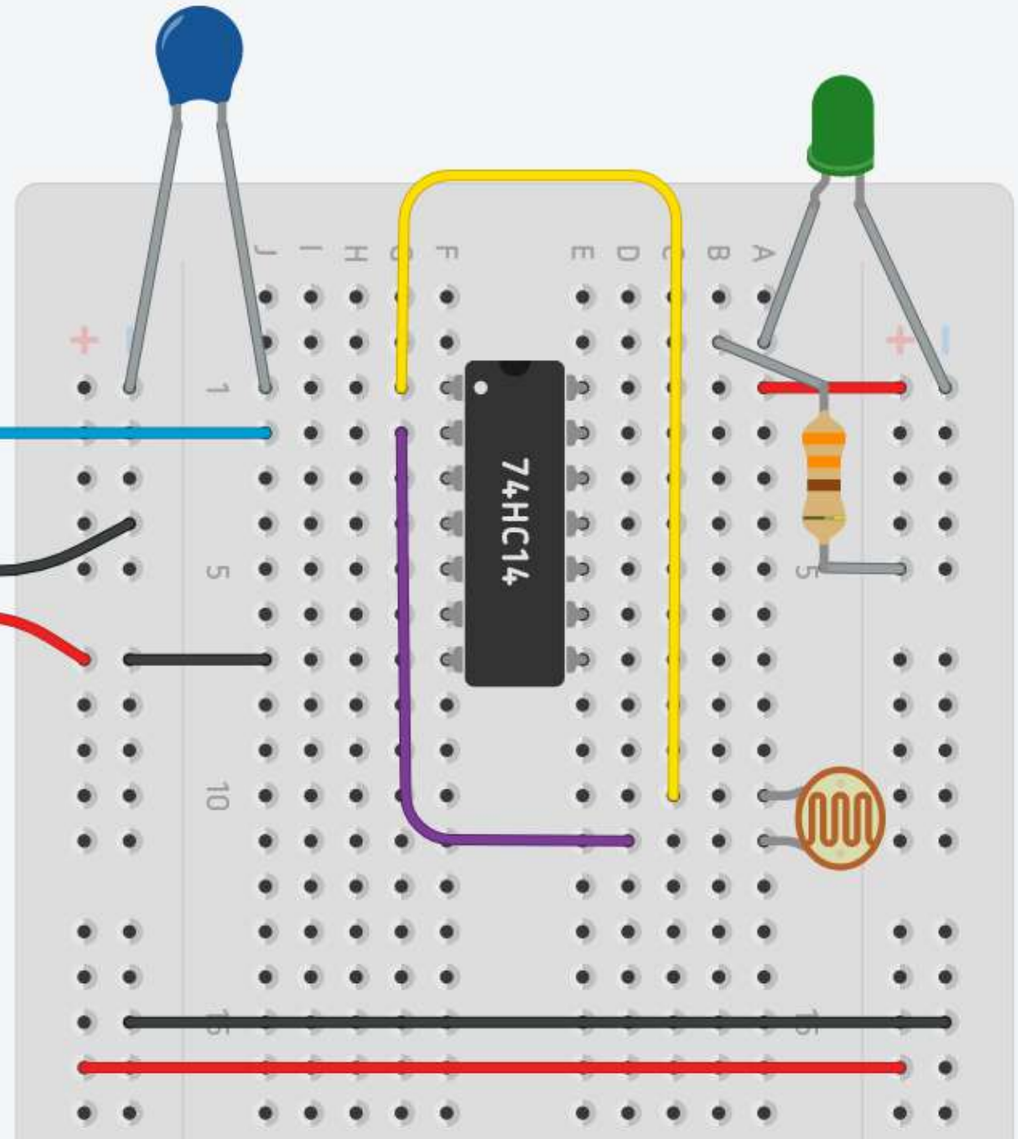




*Try different capacitor and resistor values (this LDR changes resistance based on light).

*Visualize these signals on the oscilloscope to see min and max frequencies!



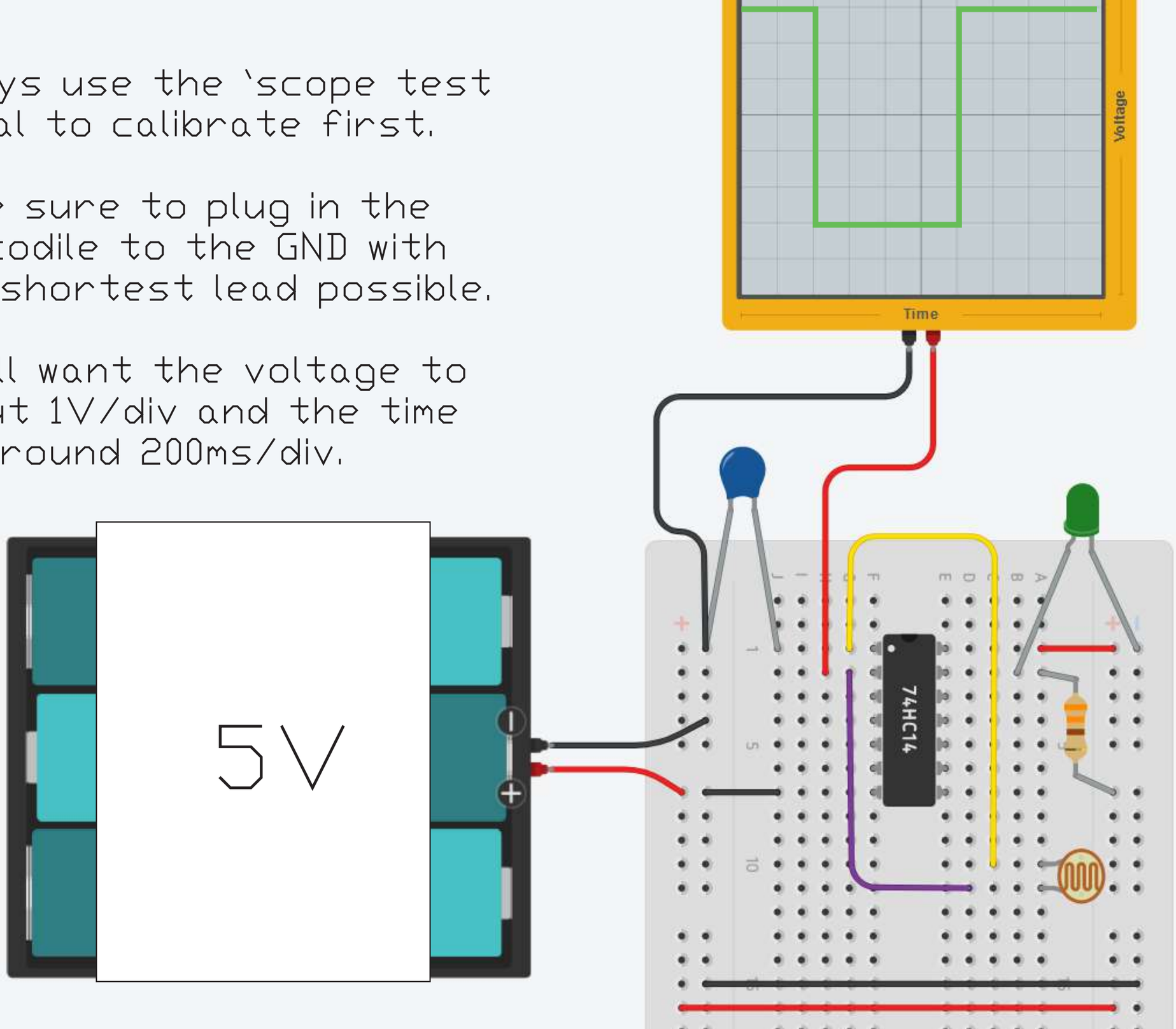


step 6

Always use the 'scope test signal to calibrate first.

Make sure to plug in the crocodile to the GND with the shortest lead possible.

You'll want the voltage to about 1V/div and the time to around 200ms/div.

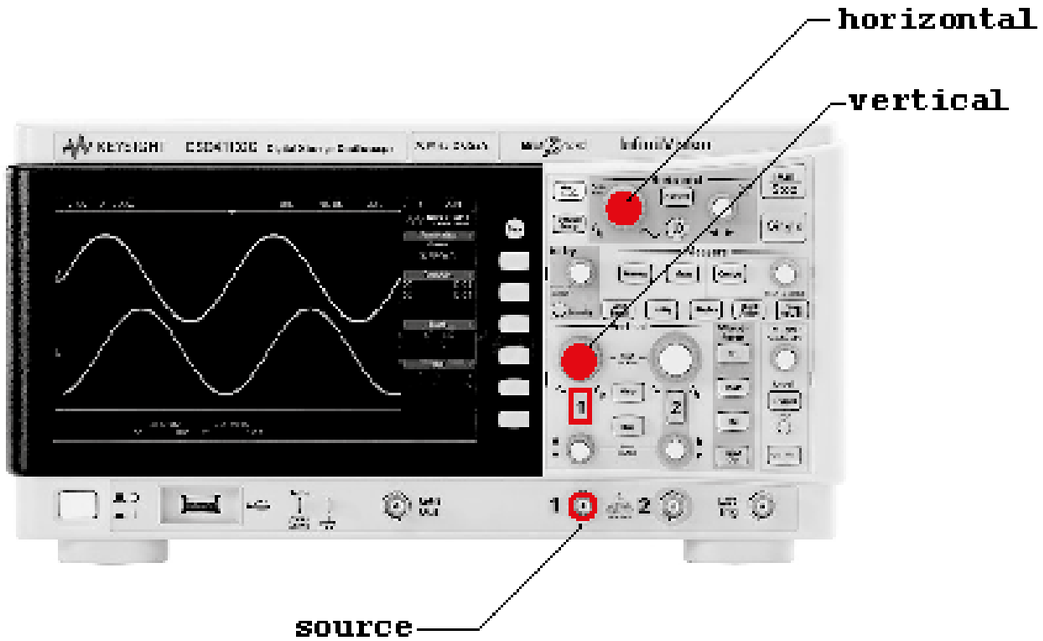


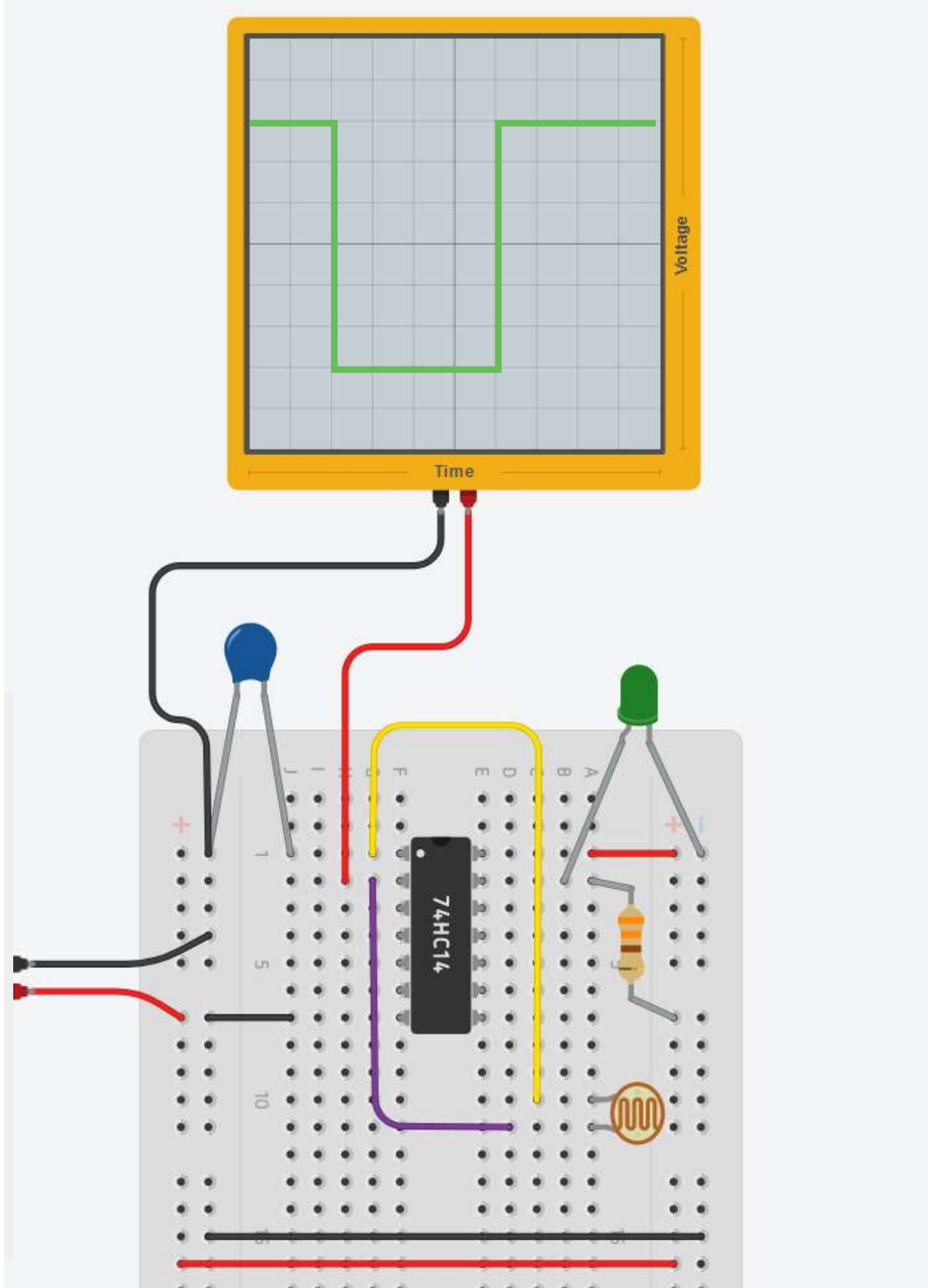
step 6

CABLE BNC



OSCILLOSCOPE

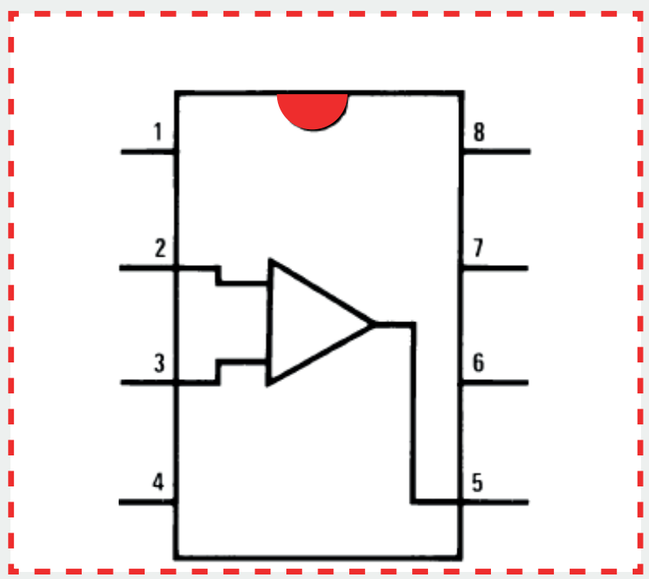


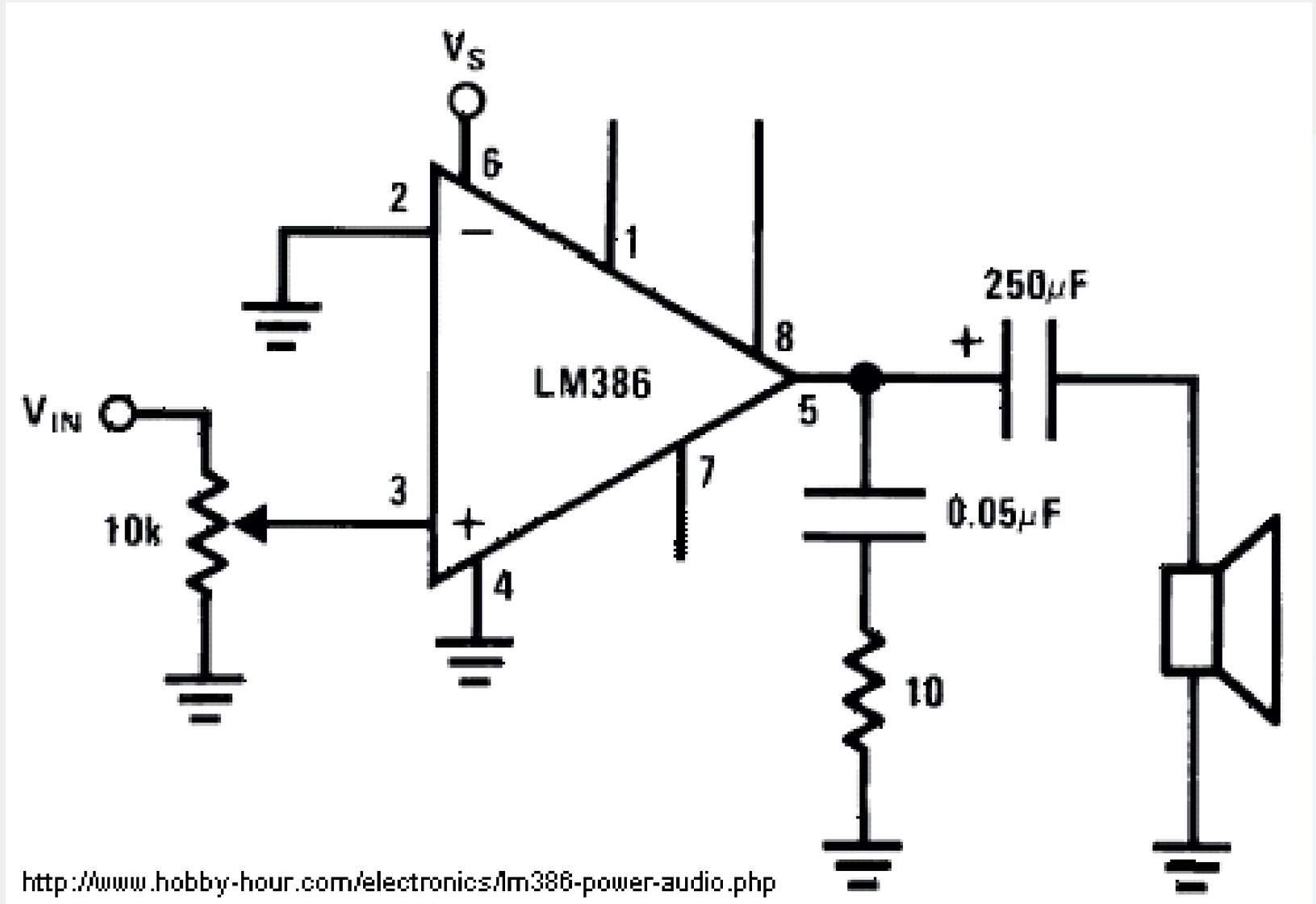


step 7

assemble an audio amp LM386 to amplify the oscillator from the 74HC14:

*notch should face north!



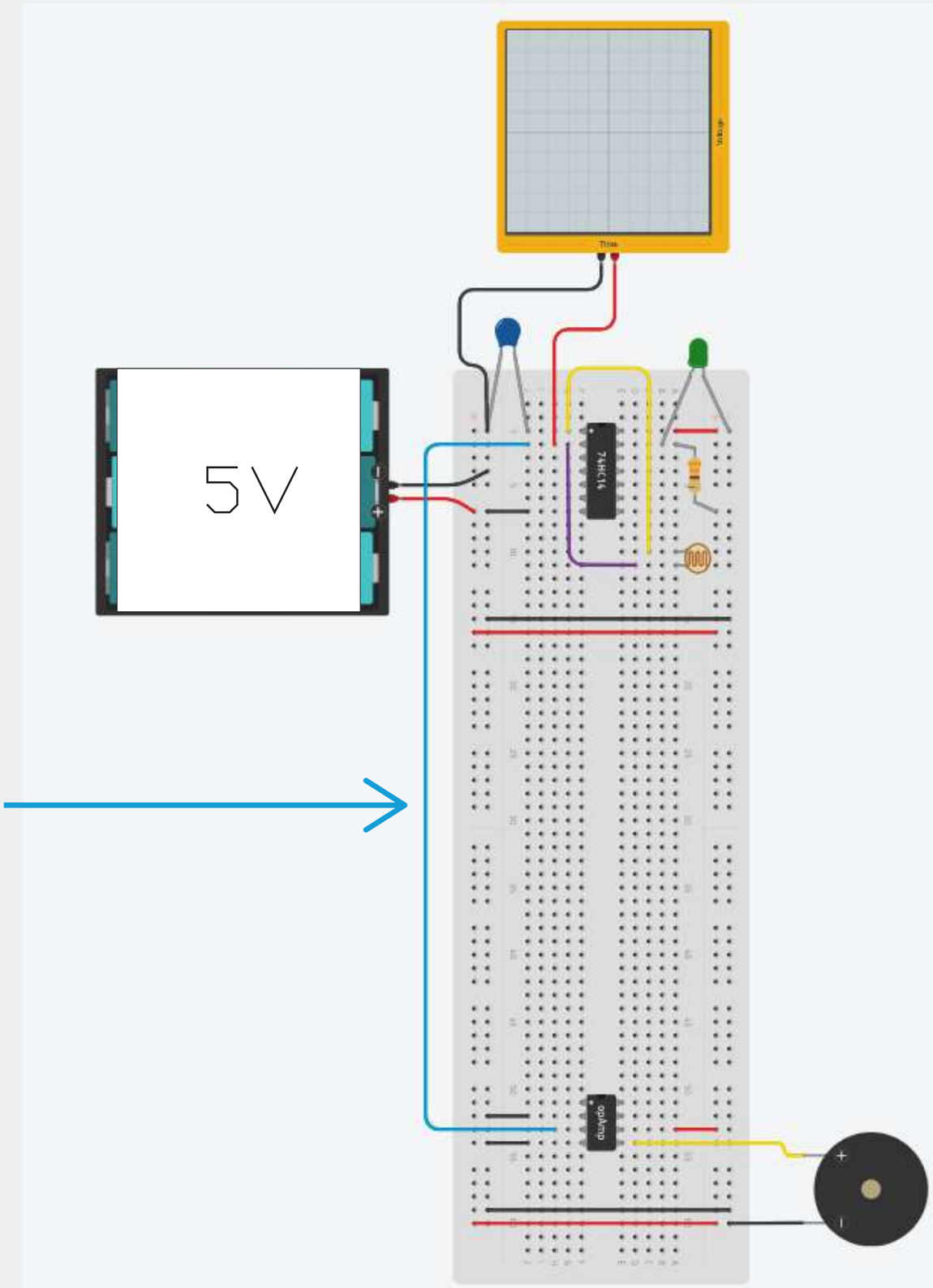


step 8

OUTPUT
(pin 2 of the
74HC14)

CONNECTED TO

INPUT
(pin 3 of the
LM386 op amp)



step 9

try mixing multiple oscillators with diodes or resistors :

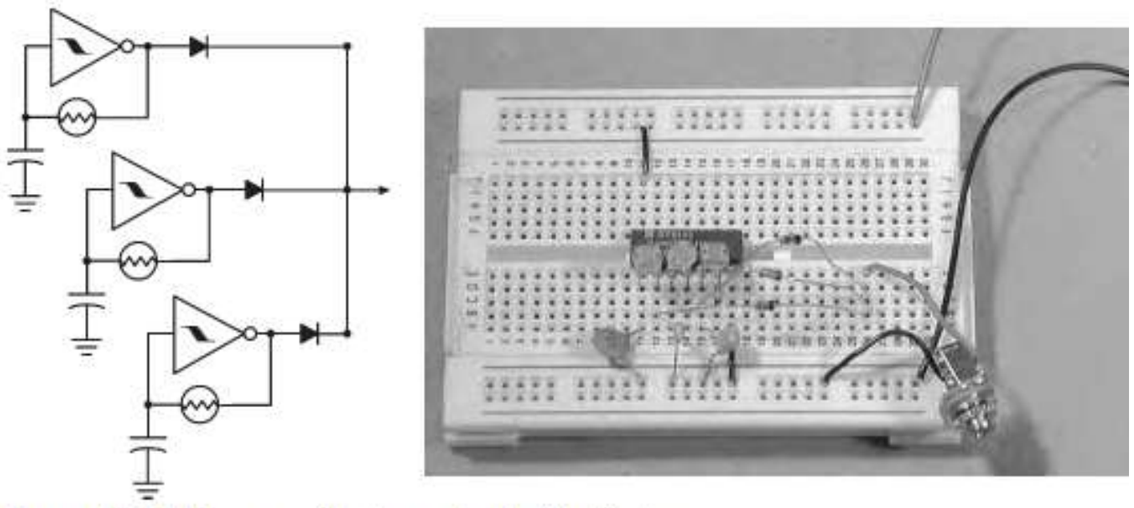


Figure 18.18 Three oscillators mixed with diodes.

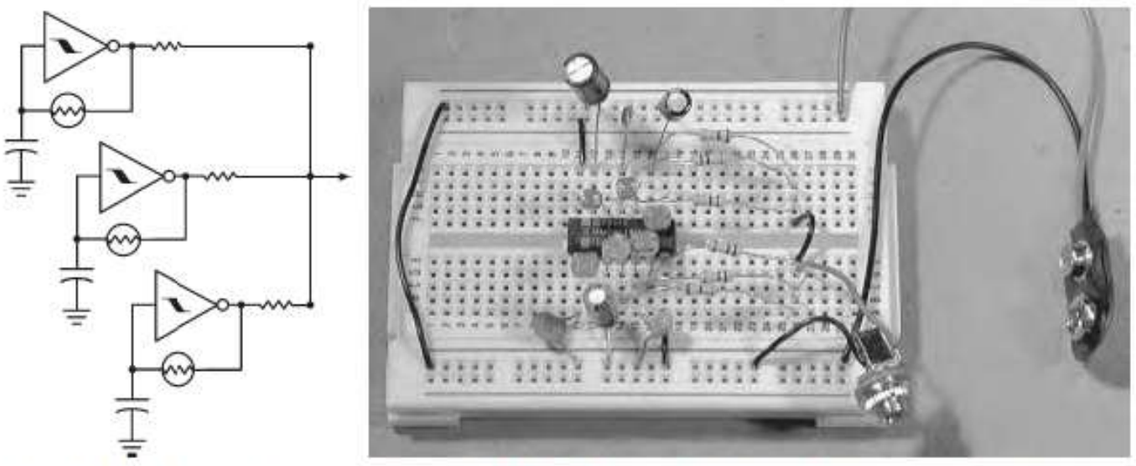


Figure 18.17 Six oscillators mixed through resistors (only three shown on schematic).

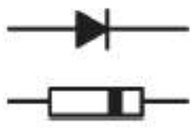


Figure 18.19 Diode orientation.

oscillator explanation :

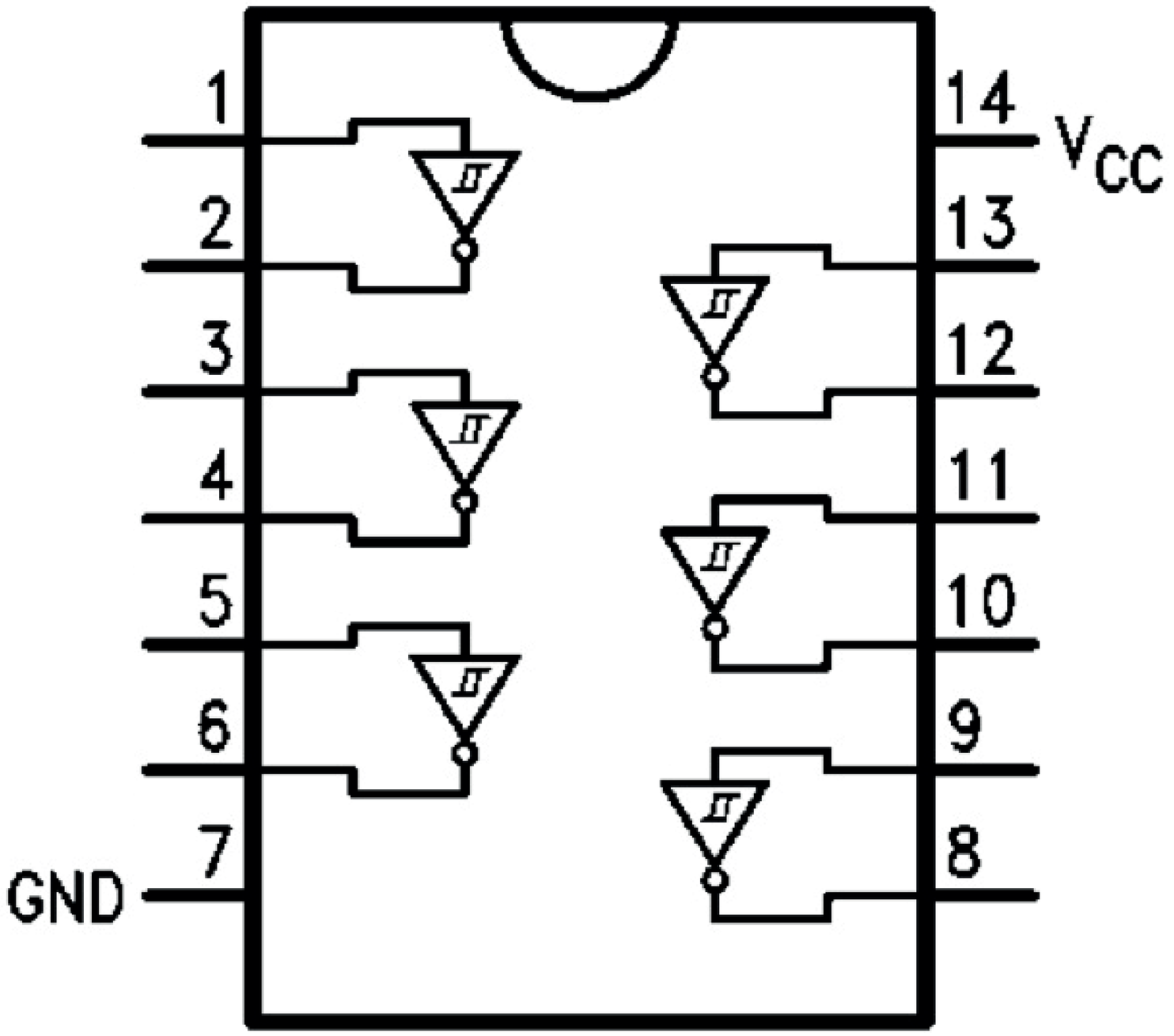
INVERTER

"YES,
I DID!"

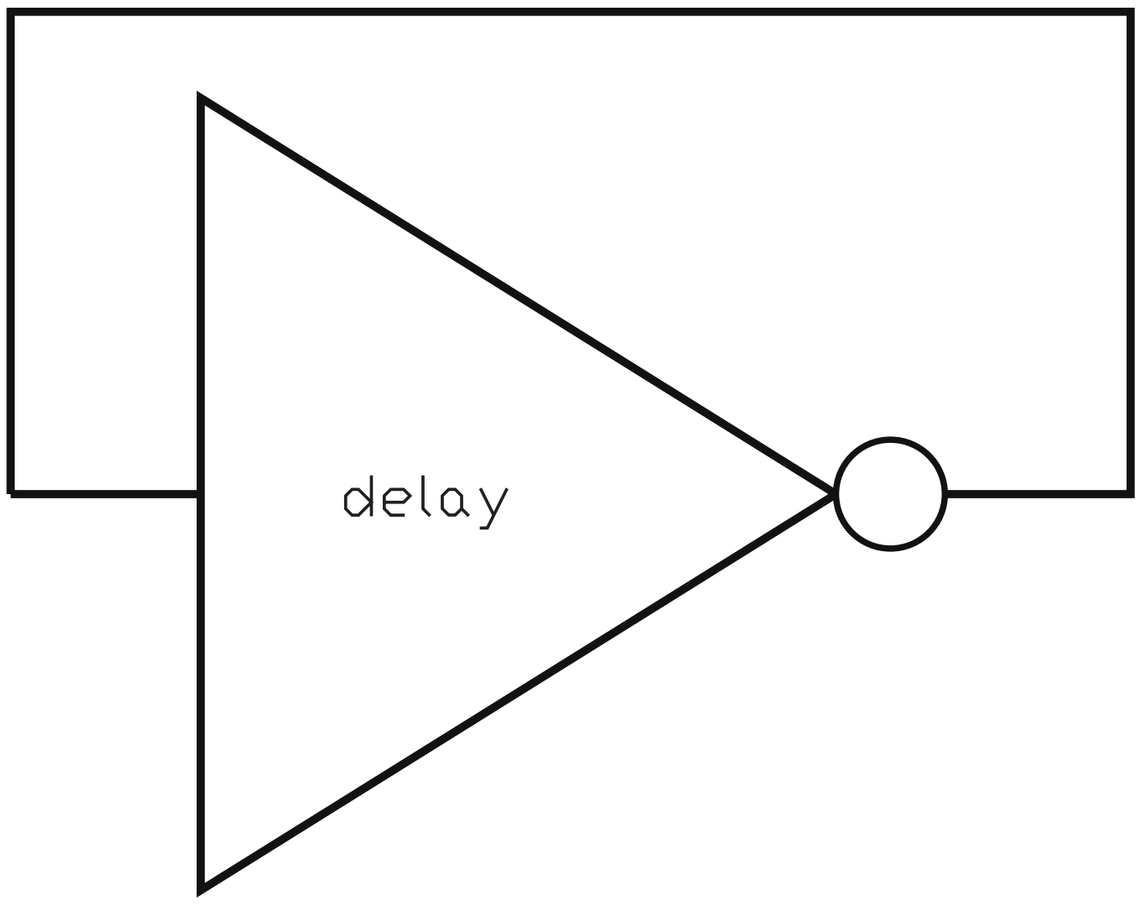


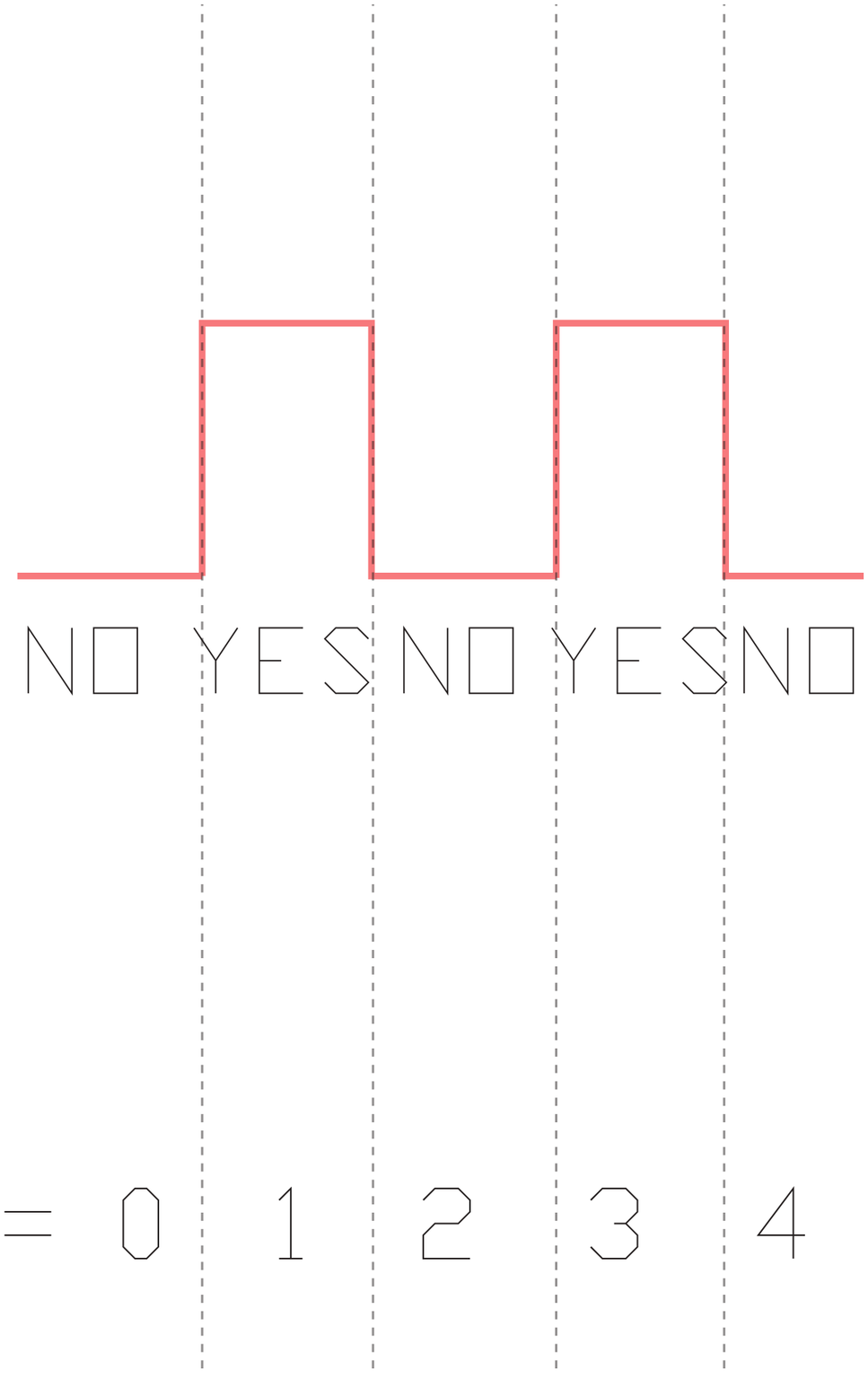
"NO,
YOU
DIDN'T"

74HC14 hex inverter

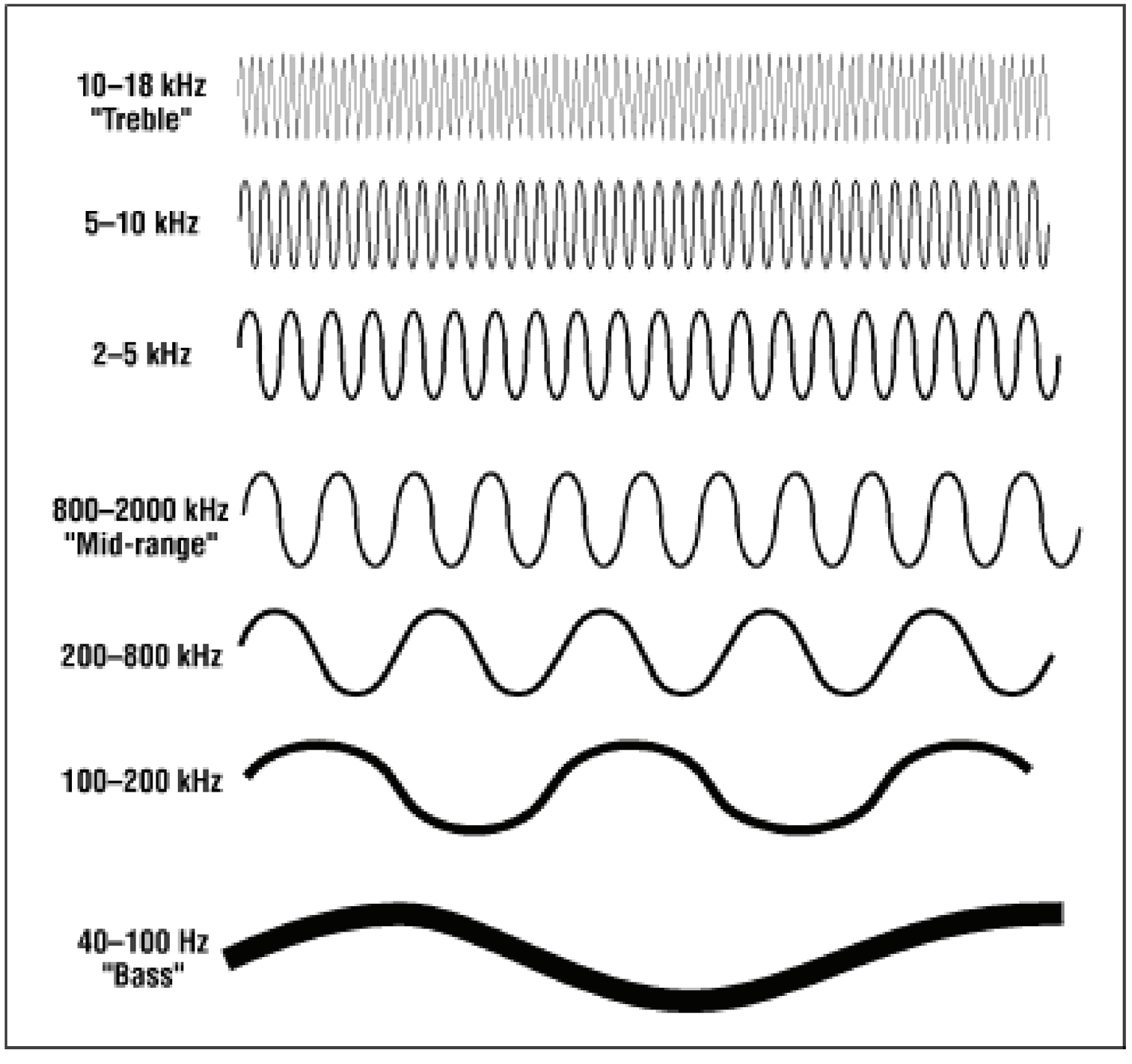


INVERTER

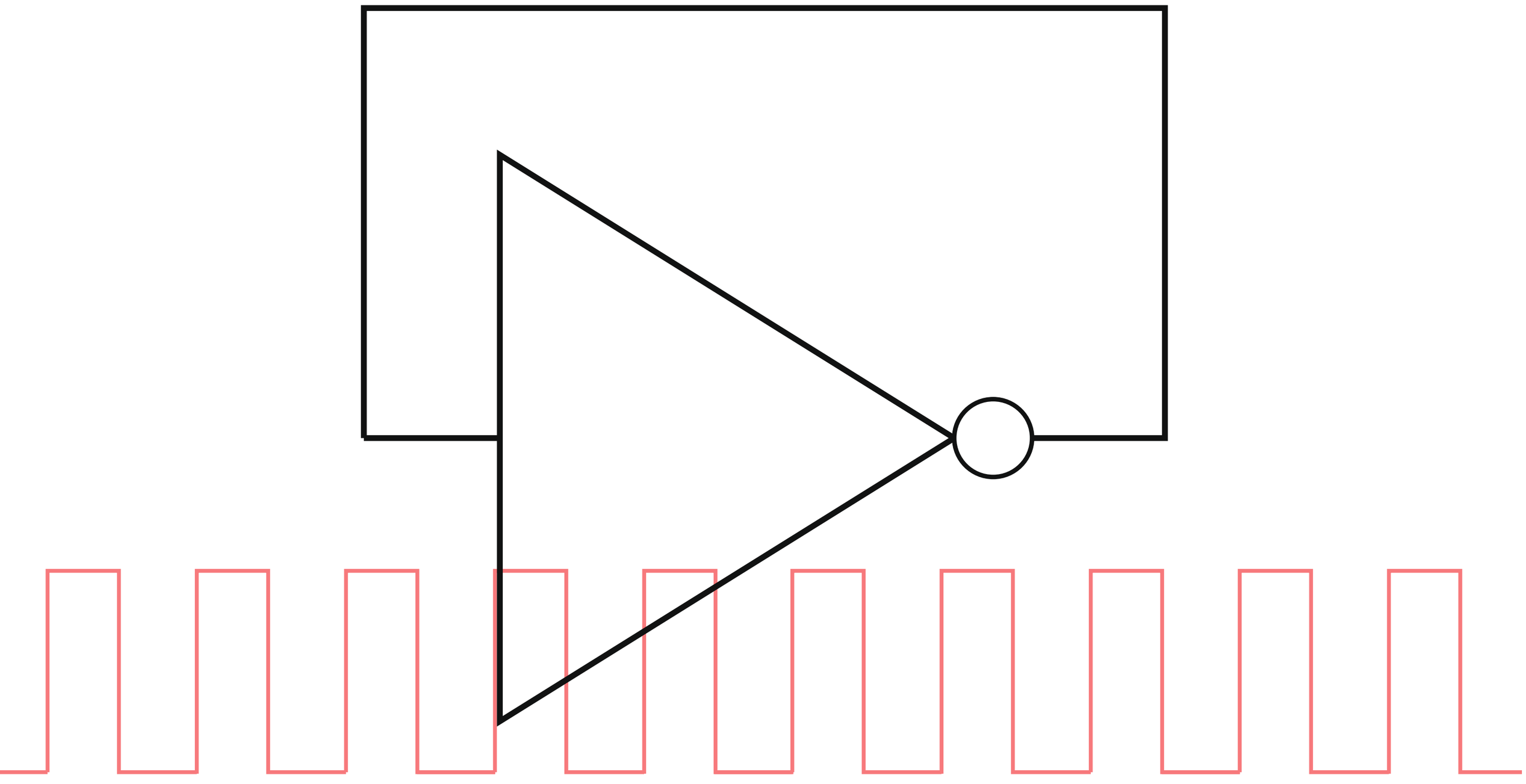




Sound = vibration !



oscillator explanation :

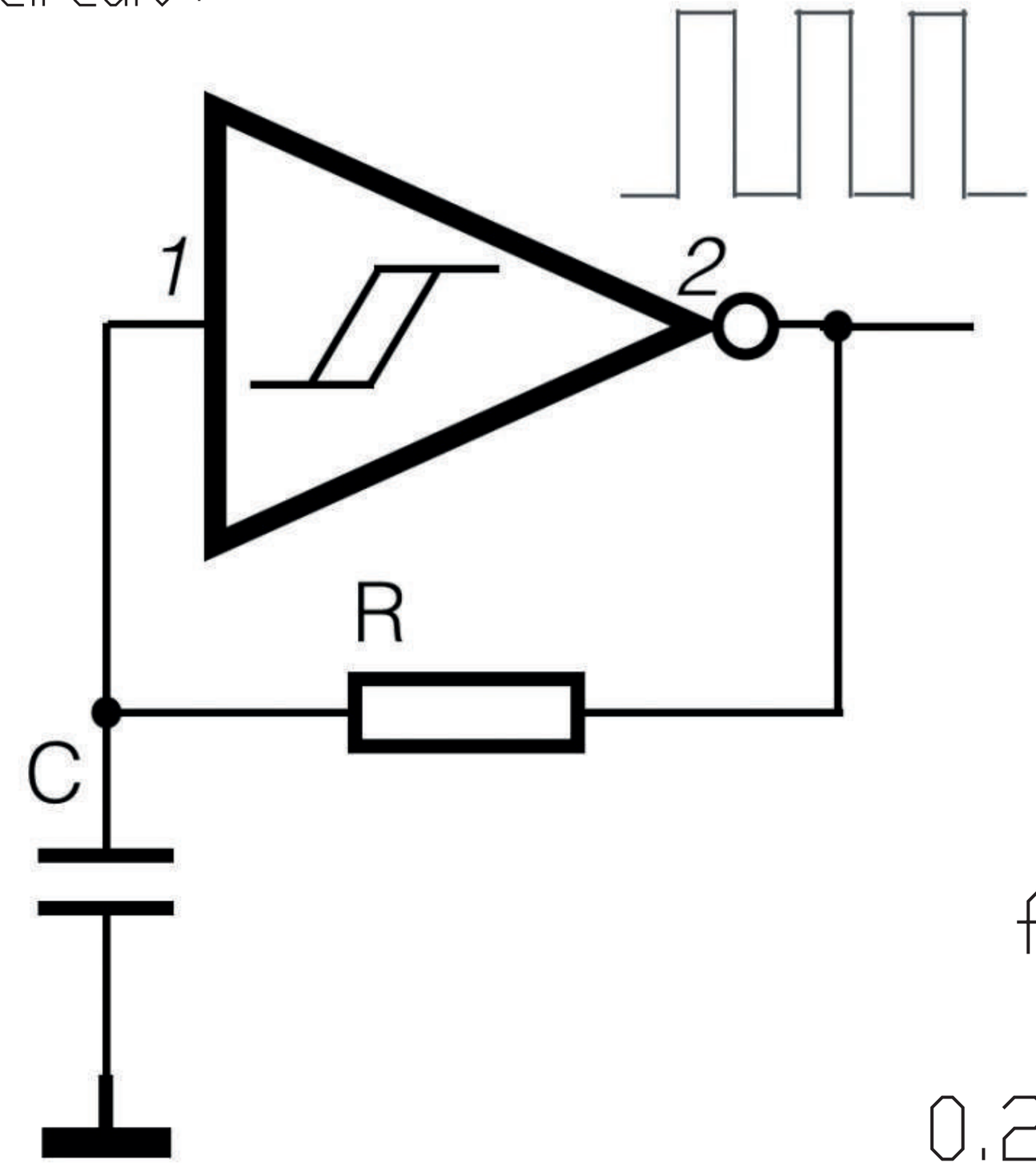


SWAMP



try to calculate the
frequency of your circuit :



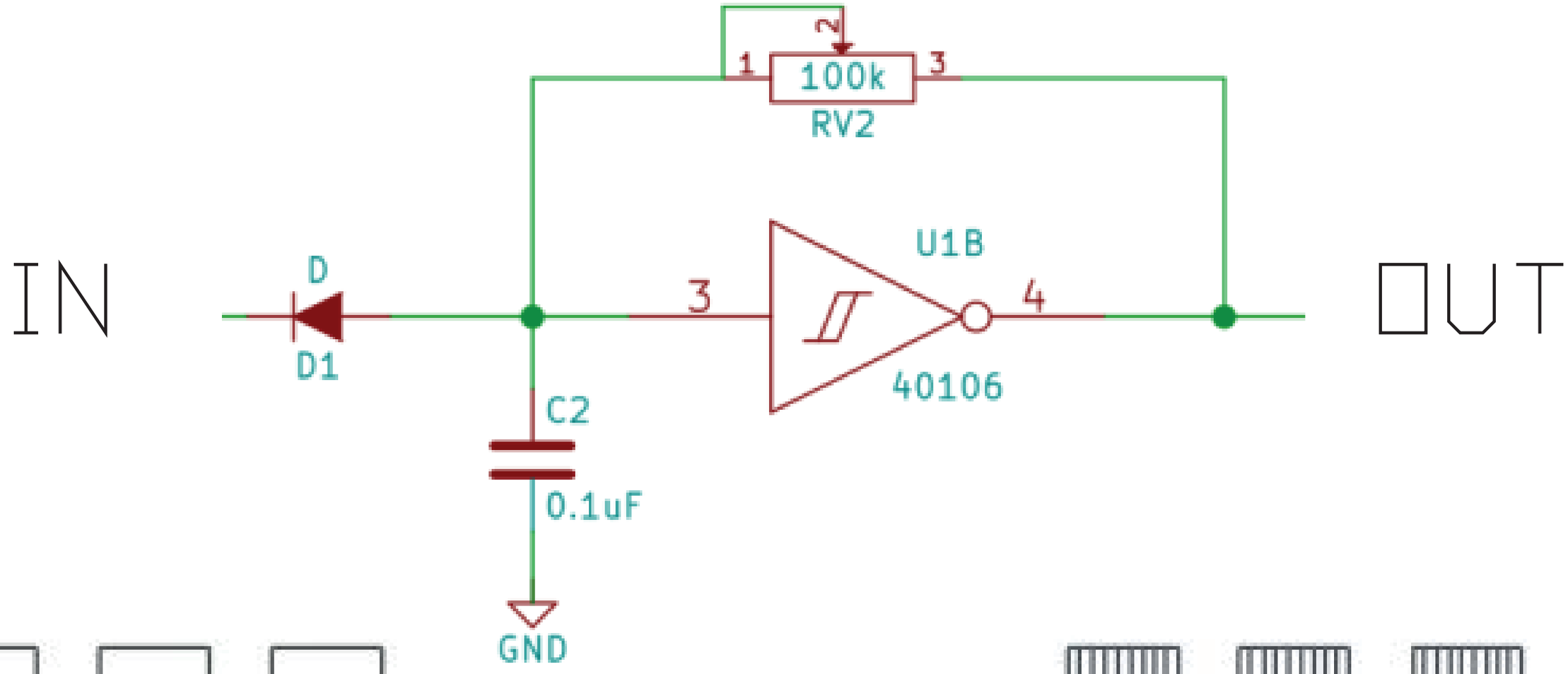


$$f = \frac{k}{RC}$$

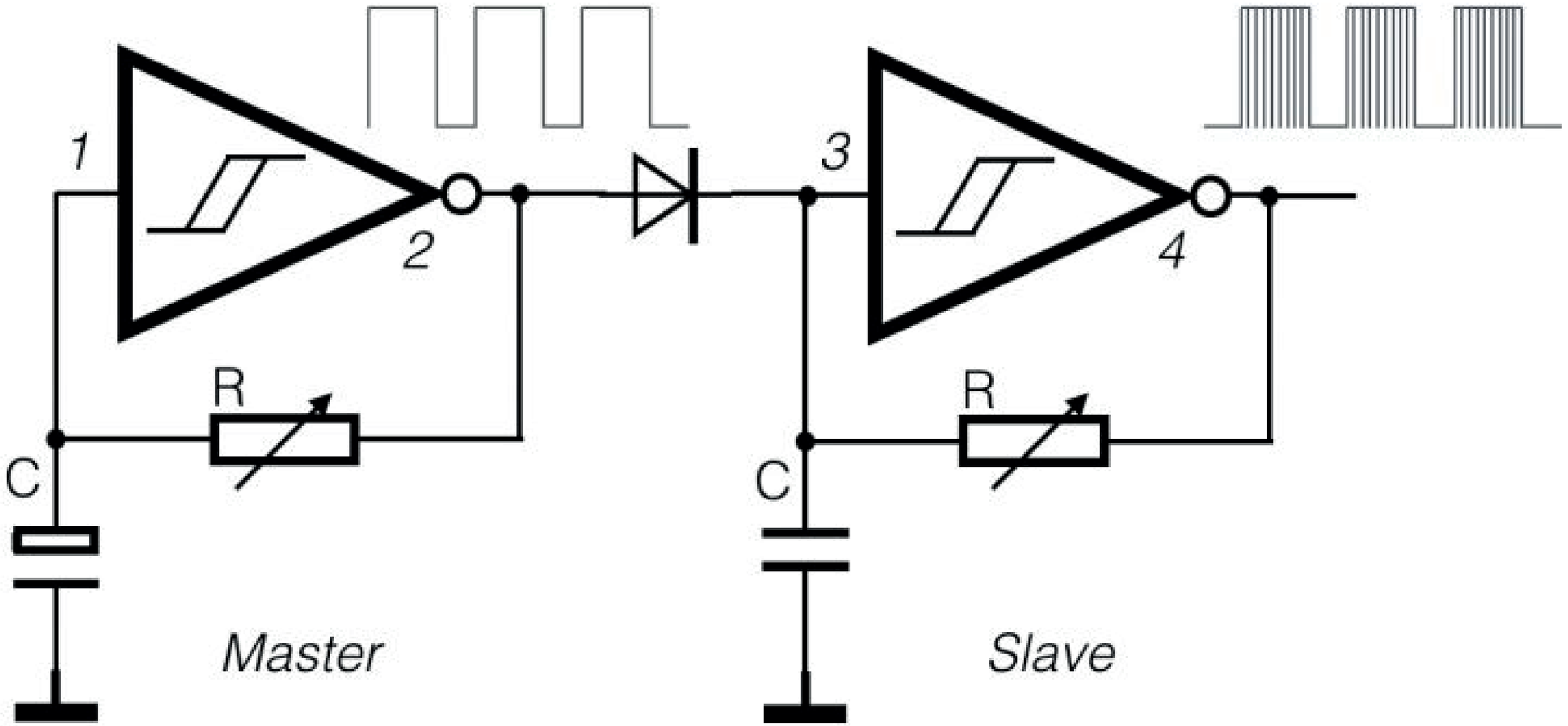
$0.2 > k < 1.0$

https://www.allaboutcircuits.com/technical-articles/exactly-how-schmitt-trigger-oscillators-work/

Could you send
satellite signals IN ?

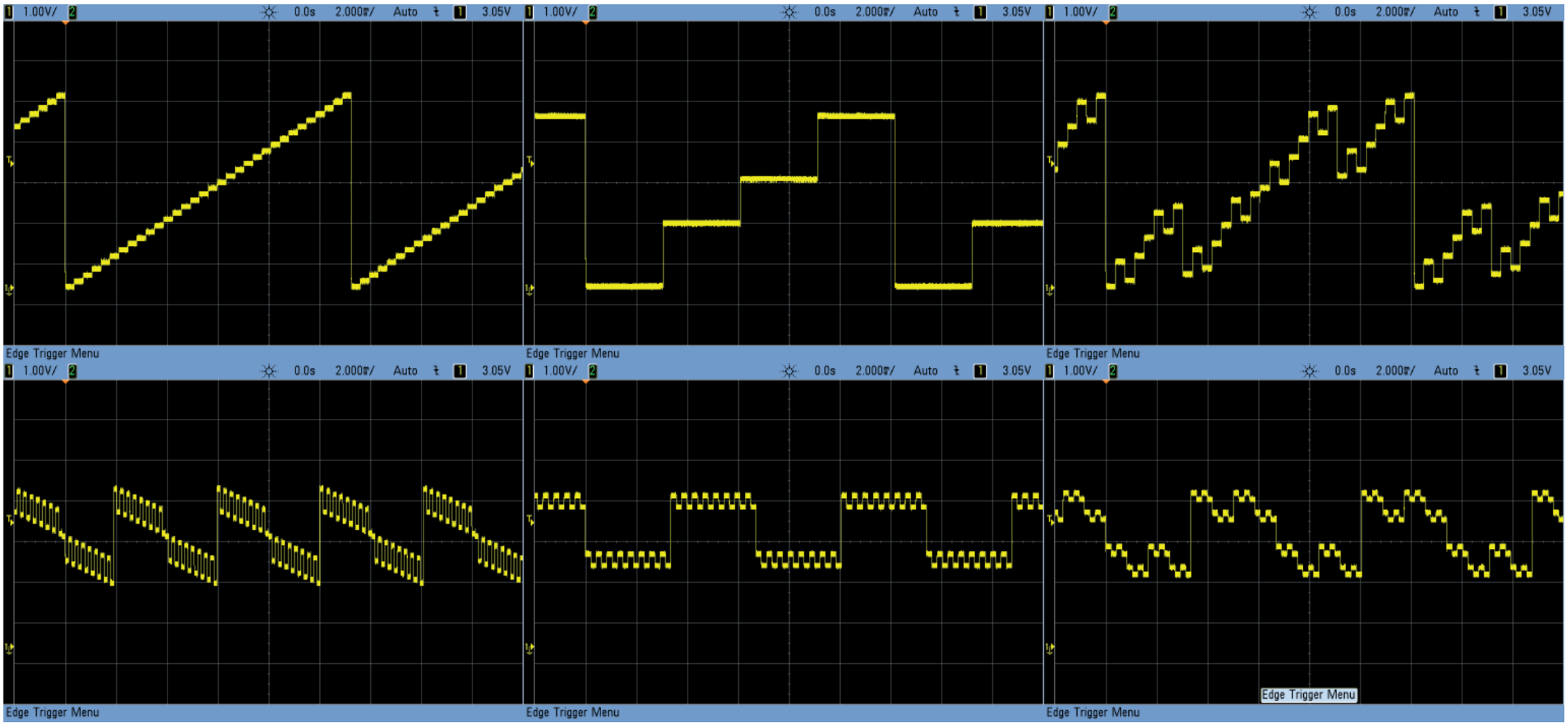


try to feed output of one oscillator into the input of another :

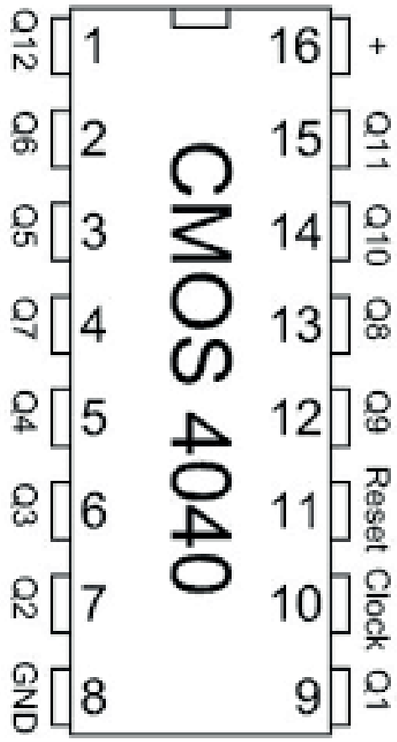


going further:

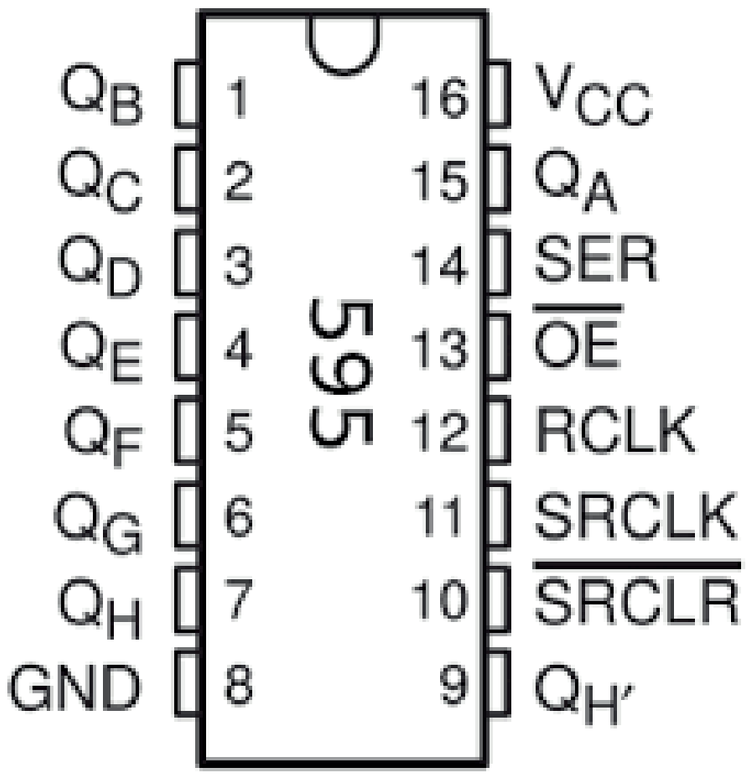
https://hackaday.com/series_of_posts/logic-noise/

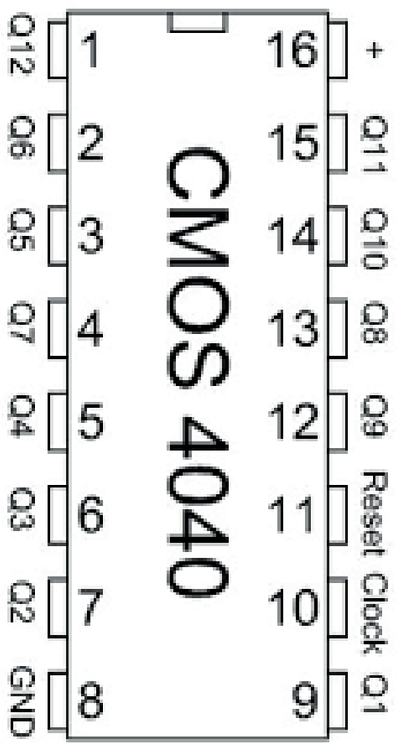


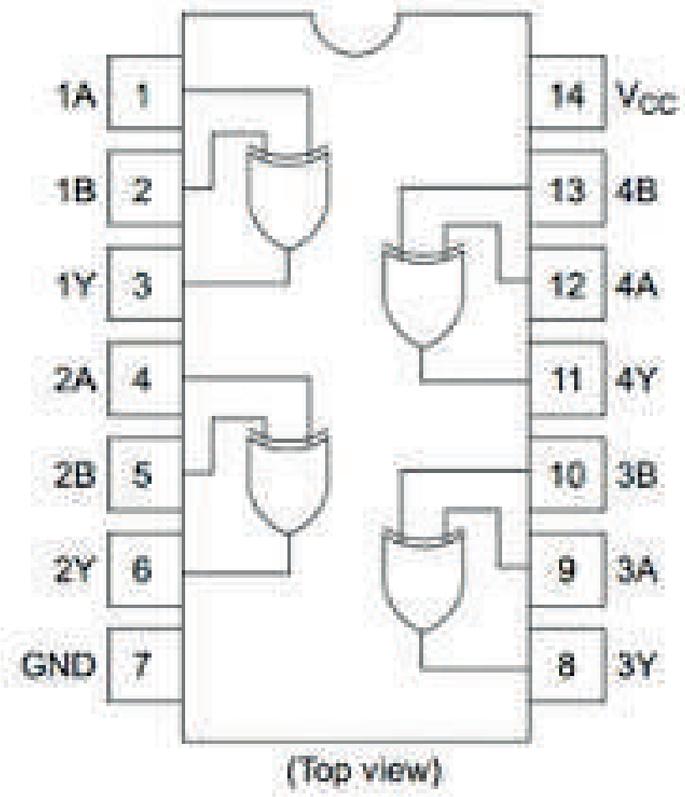
Try to feed an oscillator output into the CLOCK of the CD4040, while tying 11 HIGH.



try playing with other logic chips :







https://hackaday.com/series_of_posts/logic-noise/